Benazir Bhutto Shaheed Human Resource Research & Development Board [BBSHRRDB]
PMU-ORIC University of Sindh, Jamshoro

Practice Tasks For The Monthly Test For June 2026

Module: Version Control (Git & GitHub)

Task 1 — Initial Setup

1. Install Git and verify the version using the terminal.
2. Set your global username and email using Git config commands.
3. Verify that your configuration was saved correctly.

Task 2 — Repository Basics

1. Create a new folder called `git-practice` and initialize it as a Git repository.
2. Create a file `notes.txt` with any text inside it.
3. Check the status of your repository before and after creating the file.

Task 3 — Staging and Committing

1. Add `notes.txt` to the staging area.
2. Commit it with a meaningful commit message.
3. Make a small change to `notes.txt`, then check status again to see the difference between tracked/untracked and modified files.
4. Commit the change.

Task 4 — Working with GitHub

1. Create a new repository on GitHub (don't initialize with README).
2. Connect your local repository to the GitHub repository using the remote add command.
3. Push your commits to GitHub.
4. Make a change directly on GitHub (edit README via web interface), then pull that change to your local repo.

Task 5 — Mixed Practice

1. Create three files: `file1.txt`, `file2.txt`, `file3.txt`.
2. Add only `file1.txt` and `file2.txt` to staging (leave `file3.txt` untracked).
3. Check status to confirm.
4. Commit only the staged files.
5. Now stage and commit `file3.txt` separately.
6. Push all changes to GitHub.

Task 6 — Conceptual + Practical

1. What is the difference between `git add .` and `git add <filename>`? Demonstrate both.
2. What is the difference between `git pull` and `git fetch`? (Just answer in a comment/text file.)
3. Show the commit history of your repository using the log command.

Module: Basics of Electronics — Practice Problems

A. Ohm's Law Problems (Find I, V, or R)

1. A resistor has a resistance of 100 Ω and a voltage of 5V applied across it. Find the current flowing through it.
2. A circuit has a current of 2A flowing through a resistor of 50 Ω. Find the voltage across the resistor.
3. A voltage source of 12V causes a current of 0.5A to flow. Find the resistance of the circuit.
4. A resistor of 220 Ω has 3A of current flowing through it. Find the voltage.
5. If voltage = 9V and resistance = 3 Ω, find current.
6. A current of 0.25A flows through a resistance of 1kΩ. Find the voltage.
7. A 6V battery is connected to a resistor, and the current measured is 0.02A. Find the resistance.
8. Find the missing value:
    - V = 24V, R = 12 Ω, I = ?
    - I = 4A, R = 6 Ω, V = ?

- V = 15V, I = 3A, R = ?

B. Resistance Color Code Problems

1. Read the value of a resistor with bands: **Brown, Black, Red, Gold**.
2. Read the value of a resistor with bands: **Yellow, Violet, Orange, Gold**.
3. Read the value of a resistor with bands: **Red, Red, Green, Silver**.
4. A resistor has bands **Green, Blue, Brown, Gold** — find its resistance value and tolerance.
5. Determine the color bands for a resistor of value **4700 Ω (4.7kΩ)** with 5% tolerance.
6. Determine the color bands for a resistor of value **1 MΩ** with 5% tolerance.
7. A resistor is labeled 330 Ω ± 5%. What are its color bands?

C. Series and Parallel Combination Problems

1. Three resistors of 10 Ω, 20 Ω, and 30 Ω are connected in **series**. Find the total (equivalent) resistance.
2. Two resistors of 100 Ω and 200 Ω are connected in **parallel**. Find the equivalent resistance.
3. Four resistors of 5 Ω each are connected in **series**. Find total resistance, and if a 10V source is applied, find the total current.
4. Two resistors of 6 Ω and 3 Ω are connected in **parallel**. Find the equivalent resistance.
5. A circuit has R1 = 10 Ω and R2 = 20 Ω in series, and this series combination is connected in parallel with R3 = 15 Ω. Find total resistance of the circuit.
6. Three resistors of 4 Ω, 8 Ω, and 8 Ω are connected in **parallel**. Find total resistance.
7. If three equal resistors of 9 Ω each are connected in parallel, what is the equivalent resistance?
8. A 12V battery is connected across two resistors (4 Ω and 8 Ω) in series. Find the current flowing through the circuit and the voltage drop across each resistor.

Module: Python Practice Tasks

A. Strings and String Functions

1. Take a string input from the user and print it in uppercase, lowercase, and title case.
2. Write a program to check if a given string is a palindrome.

3. Count the number of vowels in a given string.
4. Reverse a string without using slicing (use a loop).
5. Replace all spaces in a string with underscores using a string function.
6. Take a sentence and count how many words it contains using `.split()`.
7. Check whether a string starts with "Py" and ends with "on" using string methods.

B. Conditional Statements

1. Write a program to check whether a number is positive, negative, or zero.
2. Write a program that takes marks as input and prints the grade (A, B, C, D, F) using if-elif-else.
3. Check whether a given year is a leap year.
4. Write a program to find the largest of three numbers using conditionals.
5. Take age as input and classify the person as Child, Teenager, Adult, or Senior Citizen.

C. Lists, Tuples, Dictionaries, Sets

1. Create a list of 10 numbers and find the maximum, minimum, and sum without using built-in `max()`, `min()`, `sum()` functions.
2. Write a program to remove duplicate elements from a list using a set.
3. Create a tuple of 5 fruits and try to modify one element — explain (in comments) why it gives an error.
4. Create a dictionary to store student names and their marks. Write code to add a new student, update a mark, and delete a student.
5. Given two sets, find their union, intersection, and difference.
6. Write a program to merge two dictionaries.
7. Sort a list of numbers in ascending and descending order (with and without built-in `sort()`).
8. Create a list of dictionaries representing 3 students (name, age, marks) and print each student's details using a loop.

D. Loops (while, for) and range()

1. Print numbers from 1 to 50 using a `for` loop with the `range()` function.
2. Print all even numbers between 1 and 100 using a `while` loop.
3. Write a program to print the multiplication table of a number entered by the user.
4. Write a program to find the factorial of a number using a loop.

5. Print the Fibonacci series up to 10 terms using a loop.
6. Write a program using `while` loop to keep taking user input until they type "exit".

E. Iteration over Strings, Lists, Tuples, Dicts

1. Iterate over a string and print each character along with its index.
2. Iterate over a list of numbers and print only the even numbers.
3. Iterate over a tuple of names and print each name in uppercase.
4. Iterate over a dictionary and print all key-value pairs.
5. Iterate over a dictionary and print only the keys, then only the values, using appropriate methods.

F. Type Casting

1. Take two numbers as input (input always returns string) and add them after converting to integers.
2. Convert a list to a tuple and a tuple to a list.
3. Convert a string of comma-separated numbers (e.g., "1,2,3,4") into a list of integers.
4. Convert a list of tuples into a dictionary.
5. Demonstrate type casting errors — what happens when you try `int("abc")`? (Explain in comments.)

G. Functions and Lambda Functions

1. Write a function `add(a, b)` that returns the sum of two numbers.
2. Write a function that doesn't return anything but just prints a greeting message.
3. Write a function with default parameter values (e.g., `greet(name="Guest")`).
4. Convert a normal function that squares a number into a lambda function.
5. Use `lambda` with `map()` to square all elements of a list.
6. Use `lambda` with `filter()` to filter out even numbers from a list.
7. Write a function that takes a function as a parameter and calls it (basic higher-order function example).

H. Arbitrary Arguments and **kwargs

1. Write a function `add_all(*args)` that adds any number of numbers passed to it.
2. Write a function `print_info(**kwargs)` that prints all key-value pairs passed to it (e.g., name, age, city).
3. Write a function that accepts both `*args` and `**kwargs` and prints both separately.
4. Write a function that takes a variable number of names and greets each one.

I. Recursive Functions

1. Write a recursive function to calculate the factorial of a number.
2. Write a recursive function to calculate the nth Fibonacci number.
3. Write a recursive function to find the sum of digits of a number.
4. Write a recursive function to reverse a string.

J. Local & Global Variables

1. Write a program demonstrating the difference between a local and global variable with the same name.
2. Write a function that modifies a global variable using the `global` keyword.
3. Predict the output of a given code snippet (write your own) involving local and global variable conflicts.

K. pip, File Handling, dotenv, modules, venv

1. Create a virtual environment on Windows and activate it.
2. Use `pip` to install a package (e.g., `requests`) inside the venv and verify installation using `pip list`.
3. Write a program to create a text file and write 5 lines into it.
4. Write a program to read the content of a file line by line.
5. Write a program to append new data to an existing file without overwriting it.
6. Create a `.env` file storing a fake API key and a database password, then write a Python script to read these values using `python-dotenv`.
7. Create a custom module (`mymodule.py`) with two functions, then import and use it in another script.

8. Write a program that handles a `FileNotFoundError` using try-except when trying to open a non-existent file.